

BITOLA, North Macedonia

WMO: 135830

Lat: 41.050N

Lon: 21.367E

Elev: 1932

StdP: 13.70

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.5	15.4	3.3	7.0	9.2	10.2	9.9	20.4	18.1	47.9	16.1	46.8	1.1	0	0.444

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	93.1	66.0	90.1	65.3	87.5	64.6	68.3	86.7	67.1	84.7	65.9	82.8	5.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.1	92.7	71.2	61.7	88.2	70.3	60.4	84.2	69.6	33.7	86.9	32.6	84.4	31.7	82.9	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	14.0	11.7	0.8	97.7	9.4	2.9	-6.0	99.8	-11.6	101.5	-16.9	103.1	-23.7	105.2
			1.4	70.8	8.9	1.7	-5.0	72.0	-10.2	72.9	-15.2	73.9	-21.6	75.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	32.2	37.9	45.6	52.9	60.7	67.9	72.8	72.9	64.4	54.9	45.5	35.7	(6)
	DBStd	15.42	10.16	8.52	6.81	6.50	5.25	5.48	4.79	4.26	5.90	6.86	7.26	9.03	(7)
	HDD50	1759	552	343	168	39	1	0	0	0	0	37	169	450	(8)
	HDD65	4821	1016	760	601	365	151	31	5	2	80	317	584	909	(9)
	CDD50	3110	1	3	32	125	333	537	706	711	433	188	35	6	(10)
	CDD65	699	0	0	0	2	19	118	246	249	63	2	0	0	(11)
	CDH74	8423	0	0	0	42	301	1463	2958	2841	744	74	0	0	(12)
	CDH80	3327	0	0	0	4	52	519	1292	1232	221	7	0	0	(13)
(14) Wind	WSAvg	3.8	3.1	4.0	4.9	4.9	4.0	3.8	3.9	3.5	3.6	3.1	3.4	3.1	(14)
(15) Precipitation	PrecAvg	23.7	1.9	1.7	1.8	2.2	2.2	1.4	1.2	1.3	2.2	2.6	2.5	2.8	(15)
	PrecMax	32.5	4.6	3.5	3.4	4.6	4.4	3.7	3.8	2.3	6.5	5.9	5.4	6.7	(16)
	PrecMin	15.2	0.1	0.5	0.4	0.6	0.5	0.2	0.1	0.2	0.2	0.0	0.7	0.0	(17)
	PrecStd	4.5	1.1	0.9	0.9	1.1	1.0	0.9	0.9	0.6	1.6	1.6	1.2	1.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.9	63.5	71.0	78.9	85.4	93.6	97.9	96.7	90.1	79.8	70.2	62.8	(19)
		MCWB	46.4	49.4	51.9	56.6	63.0	65.2	66.6	67.1	63.5	59.9	54.5	51.0	(20)
	2%	DB	52.2	58.0	66.0	73.1	81.1	89.2	93.2	93.2	86.2	75.7	65.1	56.1	(21)
		MCWB	44.4	46.5	49.6	55.0	60.8	64.8	66.4	66.2	62.6	57.8	53.0	48.2	(22)
	5%	DB	48.7	54.0	62.3	69.2	77.4	86.0	90.3	90.0	82.4	71.8	60.9	51.9	(23)
		MCWB	42.8	44.8	48.1	53.2	59.4	64.6	65.5	65.4	61.2	56.3	51.3	46.5	(24)
	10%	DB	45.3	50.4	58.5	65.6	74.1	82.5	87.4	87.1	78.6	68.0	57.5	48.5	(25)
		MCWB	40.7	43.2	46.5	51.7	58.1	63.0	64.8	64.5	60.1	55.0	50.4	44.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	50.5	53.3	58.4	64.7	68.6	70.0	70.2	65.6	61.6	56.8	52.6	(27)
		MCDB	54.4	60.4	65.5	74.2	80.6	85.6	90.6	89.8	83.0	74.3	63.6	59.1	(28)
	2%	WB	45.6	47.6	50.8	56.4	62.2	66.8	68.3	68.5	63.9	59.8	54.7	49.6	(29)
		MCDB	51.4	55.6	62.7	69.5	77.4	83.5	86.6	87.4	80.0	70.8	60.8	53.4	(30)
	5%	WB	43.4	45.6	49.1	54.7	60.6	65.5	67.1	67.1	62.8	58.1	53.0	47.3	(31)
		MCDB	48.3	52.6	59.8	66.2	74.3	81.5	84.7	85.0	77.5	67.1	59.2	51.4	(32)
	10%	WB	41.1	43.7	47.4	53.1	59.2	64.2	65.9	65.7	61.6	56.7	51.3	44.7	(33)
		MCDB	44.7	49.8	56.9	63.0	71.1	79.4	83.1	82.4	74.3	64.9	56.5	48.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	16.0	18.4	19.1	21.2	24.1	25.9	26.4	22.3	20.3	15.7	12.1	(35)
		MCDBR	18.0	23.4	26.8	26.8	28.3	30.4	31.5	31.3	30.4	28.3	23.3	16.8	(36)
	5% WB	MCWBR	10.9	12.8	12.6	11.3	10.5	9.5	8.9	8.9	9.8	11.8	12.4	9.9	(37)
		MCWBR	16.5	20.6	23.4	23.4	25.3	27.5	28.0	27.9	24.8	22.3	17.7	13.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.359	0.408	0.470	0.463	0.438	0.442	0.457	0.442	0.398	0.359	0.331	(40)
		taud	2.347	2.250	2.143	1.985	2.069	2.191	2.179	2.131	2.142	2.249	2.320	2.337	(41)
	Ebn at Noon		258	266	264	255	260	267	264	256	250	248	243	246	(42)
		Edh at Noon	27	35	43	55	52	46	46	47	43	35	28	26	(43)
(44) All-Sky Solar Radiation	RadAvg		556	819	1151	1485	1797	2074	2174	1933	1415	994	648	484	(44)
	RadStd		80	104	140	169	120	154	124	122	120	109	66	86	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (19 neighbors)		+0.76	N/A	N/A	N/A	N/A	N/A	N/A	-209	N/A	N/A	

Nomenclature: See separate page